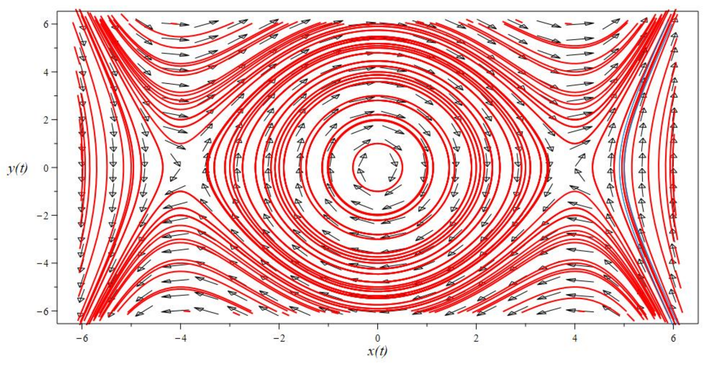
Ollscoil na Gaillimhe
University of Galway

# Undergraduate Research

## at the School of Mathematical and Statistical Sciences, University of Galway

This poster presents a small selection of the images created by undergraduate students in the School of Mathematics, Statistics and Applied Mathematics, as part of their final year projects. They show the diversity of topics our students are researching, covering numerous areas of pure and applied mathematics, statistics, bioinformatics, and computing. For more information, use the QR-code above.

### Thomas O'Connor: *The Duffing Equation*




### Dara Kelly and Cronan Ward: *Modelling Extremes in Finance and Investigating Asymptotic Dependence*

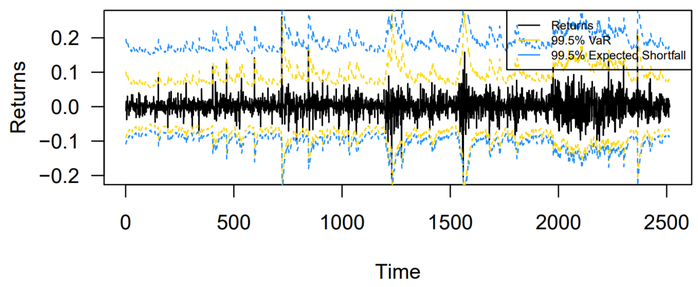
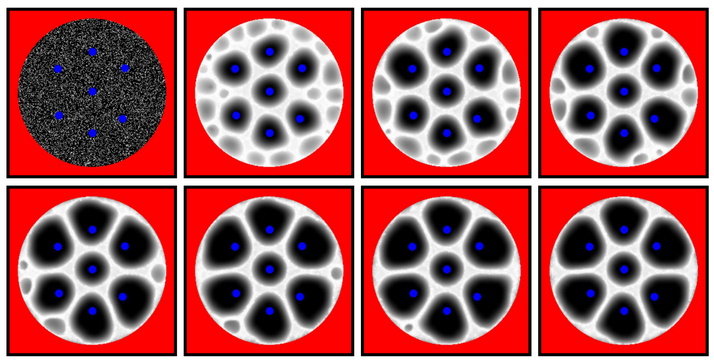
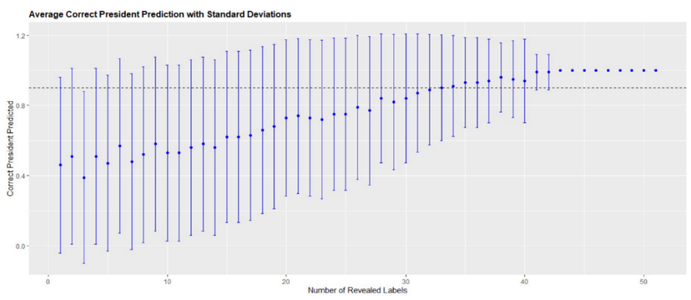



Figure 25: NVDA returns with VaR 99.5% and Expected Shortfall 99.5% Bounds (1st Jan 2014−2024)

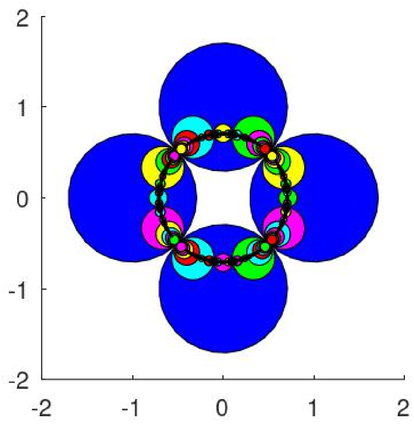
### Sean Janson and Dmytro Lyubka: *A title would be nice*



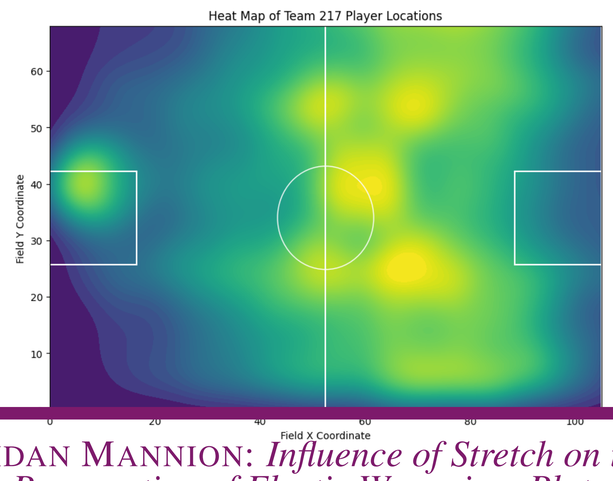
### Lise Wall and Emma Meaney: *Predicting the US president election using network approaches*




### Andrew Bell and Shane Cotter: *Fractals and Mobius Maps*



### Ailbhe Keane, Karen Gillooly and Paulina Krungleviciute: *Analysis of Expected Goals (xG) of Team 217 during La Liga 2019/2020 season*




### Aaron Kilboy: *An Investigation into Bitcoin and Blockchain Technology*

### Pierce Lawlor: *Optimal Design of a Ski Ramp*

### Aidan Mannion: *Influence of Stretch on the Propagation of Elastic Waves in a Plate*

### Lorna McLoughlin: *The effects of the IGF1 gene on cancer risk*

### Noeleen Rawle: *High dimensional quadrature using sparse grid methods*

### Brian Regan: *Markov Chains and an Application in Modeling and Simulating Basketball Games*

### David Smyth: *A Statistical Analysis and Visualization of High Performance Athlete Data*

### Szymon Urbas: *Identification and evaluation of structural equation models*

### Bartlomiej Zaucha: *Dancing links*